

TCS NQT

Numerical Ability Formula Handbook

Every formula, rule and quick shortcut across all 22 Numerical Ability topics — organized for fast revision before the test.

22 topics · Clocks to Standard Deviation · Quick-reference format

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How this handbook is organized

Each topic opens with a navy banner, then groups its formulas under short subheadings. Bold text names the concept; the formula itself is set in monospace amber-brown so it's easy to spot on the page while skimming. Italic notes under a group call out a shortcut, edge case, or the quickest way to apply the rule in an exam setting.

01 Clocks & Calendar

CLOCK — FUNDAMENTALS

- ▶ Angle traced by minute hand: 6° per minute
- ▶ Angle traced by hour hand: 0.5° per minute ($\frac{1}{2}^\circ/\text{min}$)
- ▶ Relative speed of hands: 5.5° per minute ($11/2^\circ/\text{min}$)
- ▶ Angle between hands at H hrs M min: $\theta = |30H - 5.5M|^\circ$ (if $\theta > 180^\circ$, use $360^\circ - \theta$)

CLOCK — COINCIDENCE, OPPOSITE & RIGHT ANGLE

- ▶ Hands coincide (0°): Every $65(5/11)$ min $\rightarrow$ $11 \times / 12$ hrs, $22 \times / 24$ hrs
- ▶ Hands opposite (180°): Every $65(5/11)$ min $\rightarrow$ $11 \times / 12$ hrs, $22 \times / 24$ hrs
- ▶ Hands at right angle (90°): $22 \times / 12$ hrs, $44 \times / 24$ hrs
- ▶ Minutes past H o'clock for angle θ : $M = (2/11) \times (30H \pm \theta)$
- ▶ Faulty clock comparison: (time shown by faulty clock) / (true elapsed time) = constant ratio

CALENDAR — ODD DAYS

- ▶ Ordinary year: $365 \text{ days} = 52 \text{ weeks} + 1 \text{ day} \rightarrow 1 \text{ odd day}$
- ▶ Leap year: $366 \text{ days} = 52 \text{ weeks} + 2 \text{ days} \rightarrow 2 \text{ odd days}$
- ▶ Leap year test: Divisible by 4 and NOT a century year, OR divisible by 400
- ▶ 100 / 200 / 300 / 400 years: 5 / 3 / 1 / 0 odd days respectively
- ▶ Odd days $\rightarrow$ weekday code: 0=Sun 1=Mon 2=Tue 3=Wed 4=Thu 5=Fri 6=Sat

Add up odd days for the full period, take mod 7, then map to the weekday code.

02 Series and Progressions

ARITHMETIC PROGRESSION (AP)

- ▶ nth term: $a_n = a + (n - 1)d$
- ▶ Sum of n terms: $S_n = n/2 \times [2a + (n - 1)d] = n/2 \times (a + l)$
- ▶ Common difference: $d = a_n - a_{n-1}$
- ▶ Arithmetic mean of a, b: $(a + b) / 2$

GEOMETRIC PROGRESSION (GP)

- ▶ nth term: $a_n = a \cdot r^{(n - 1)}$
- ▶ Sum of n terms ($r \neq 1$): $S_n = a(r^n - 1)/(r - 1)$ [$r > 1$] or $a(1 - r^n)/(1 - r)$ [$r < 1$]
- ▶ Sum to infinity ($|r| < 1$): $S_\infty = a / (1 - r)$
- ▶ Geometric mean of a, b: $\sqrt{ab}$

HARMONIC PROGRESSION (HP)

- ▶ Definition: Reciprocals of an HP form an AP
- ▶ nth term of HP: $1 / (\text{nth term of the reciprocal AP})$
- ▶ Harmonic mean of a, b: $2ab / (a + b)$
- ▶ AM-GM-HM relation: $AM \geq GM \geq HM$ (equal only when all terms are equal)

SPECIAL SUMMATION FORMULAS

- ▶ Sum of first n natural numbers: $n(n + 1) / 2$
- ▶ Sum of squares (1^2 to n^2): $n(n + 1)(2n + 1) / 6$
- ▶ Sum of cubes (1^3 to n^3): $[n(n + 1) / 2]^2$
- ▶ Sum of first n odd numbers: n^2
- ▶ Sum of first n even numbers: $n(n + 1)$

03 Equations

LINEAR EQUATIONS

- ▶ One variable: $ax + b = 0$: $x = -b / a$
- ▶ Two variables: Solve by elimination or substitution; unique solution needs $a_1/a_2 \neq b_1/b_2$

QUADRATIC EQUATIONS

- ▶ Standard form: $ax^2 + bx + c = 0$ ($a \neq 0$)
- ▶ Roots (quadratic formula): $x = [-b \pm \sqrt{b^2 - 4ac}] / 2a$
- ▶ Sum of roots: $\alpha + \beta = -b / a$
- ▶ Product of roots: $\alpha \cdot \beta = c / a$
- ▶ Discriminant $D = b^2 - 4ac$: $D > 0$: real, distinct roots · $D = 0$: real, equal roots · $D < 0$: complex roots
- ▶ Equation from given roots: $x^2 - (\text{sum of roots})x + (\text{product of roots}) = 0$

04 Averages

BASICS

- ▶ Average: $\text{Average} = (\text{Sum of observations}) / (\text{Number of observations})$
- ▶ Sum from average: $\text{Sum} = \text{Average} \times \text{Number of observations}$

COMBINED & WEIGHTED AVERAGE

- ▶ Combined average of two groups: $(n_1a_1 + n_2a_2) / (n_1 + n_2)$
- ▶ Weighted average: $(w_1x_1 + w_2x_2 + \dots + w_nx_n) / (w_1 + w_2 + \dots + w_n)$

AVERAGE SPEED

- ▶ Equal distances, speeds a & b: $2ab / (a + b)$
- ▶ Equal times, speeds a & b: $(a + b) / 2$

CHANGE-IN-AVERAGE SHORTCUTS

- ▶ One value replaced: $\text{New average} = \text{Old average} + (\text{New value} - \text{Old value}) / n$
- ▶ New member included: $\text{New average} = (n \times \text{Old average} + \text{New value}) / (n + 1)$

05 Area, Shapes & Perimeter

QUADRILATERALS

- ▶ Square: Area = a^2 · Perimeter = $4a$ · Diagonal = $a\sqrt{2}$
- ▶ Rectangle: Area = $l \times b$ · Perimeter = $2(l + b)$ · Diagonal = $\sqrt{l^2 + b^2}$
- ▶ Parallelogram: Area = base $\times$ height
- ▶ Rhombus: Area = $\frac{1}{2} \times d_1 \times d_2$ · Perimeter = $4 \times \text{side}$
- ▶ Trapezium: Area = $\frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$

TRIANGLES

- ▶ General: Area = $\frac{1}{2} \times \text{base} \times \text{height}$
- ▶ Heron's formula: Area = $\sqrt{s(s - a)(s - b)(s - c)}$, $s = (a + b + c)/2$
- ▶ Equilateral triangle: Area = $(\sqrt{3}/4) a^2$ · Height = $(\sqrt{3}/2) a$
- ▶ Right triangle: Area = $\frac{1}{2} \times (\text{product of the two legs})$

CIRCLE & SECTOR

- ▶ Circle: Area = πr^2 · Circumference = $2\pi r$
- ▶ Sector area: $(\theta / 360^\circ) \times \pi r^2$
- ▶ Arc length: $(\theta / 360^\circ) \times 2\pi r$
- ▶ Ring / annulus area: $\pi(R^2 - r^2)$

06 Numbers & Decimal Fractions

FRACTIONS

- ▶ Fraction: Numerator / Denominator
- ▶ Proper vs improper: Proper: Numerator < Denominator • Improper: Numerator $\geq$ Denominator
- ▶ Comparing fractions: Cross-multiply, or convert to a common denominator

DECIMALS

- ▶ Decimal fraction: A fraction whose denominator is a power of 10
- ▶ Trailing zeros: Annexing zeros after the last decimal digit does not change the value

RECURRING DECIMALS → FRACTION

- ▶ Pure recurring: $0.ababab\dots = ab / 99$ (repeating block over as many 9s)
- ▶ Mixed recurring: (all digits as a number – non-repeating digits as a number) / (9s for repeating digits, followed by 0s for non-repeating digits)

HANDY IDENTITIES

- ▶ Square of sum: $(a + b)^2 = a^2 + 2ab + b^2$
- ▶ Square of difference: $(a - b)^2 = a^2 - 2ab + b^2$
- ▶ Difference of squares: $a^2 - b^2 = (a + b)(a - b)$

07 Number System

CLASSIFICATION

- ▶ Natural / Whole numbers: $N = \{1, 2, 3, \dots\}$ $W = \{0, 1, 2, 3, \dots\}$
- ▶ Integers: $Z = \{\dots, -2, -1, 0, 1, 2, \dots\}$
- ▶ Rational / Irrational: Rational = p/q form ($q \neq 0$) · Irrational cannot be written as p/q

PLACE & FACE VALUE

- ▶ Face value: The digit itself
- ▶ Place value: Digit $\times$ (10 raised to its positional power)

USEFUL IDENTITIES

- ▶ Sum of cubes: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- ▶ Difference of cubes: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
- ▶ Square of a trinomial: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

QUICK TESTS

- ▶ Perfect square — last digit: Must end in 0, 1, 4, 5, 6 or 9 (never 2, 3, 7, 8)
- ▶ Number of factors: If $N = p_1^a \times p_2^b \times p_3^c \dots$, factor count = $(a+1)(b+1)(c+1)\dots$

08 LCM & HCF

CORE RELATION

- ▶ Two numbers: $\text{Product of the two numbers} = \text{LCM} \times \text{HCF}$

METHODS

- ▶ Prime factorization: $\text{HCF} = \text{product of lowest powers of common primes}$ · $\text{LCM} = \text{product of highest powers of all primes present}$
- ▶ Division method: Repeated division (Euclid's algorithm) to find HCF

FRACTIONS

- ▶ HCF of fractions: $\text{HCF}(\text{numerators}) / \text{LCM}(\text{denominators})$
- ▶ LCM of fractions: $\text{LCM}(\text{numerators}) / \text{HCF}(\text{denominators})$

APPLICATIONS

- ▶ Same remainder r for x, y, z : $\text{Required number} = \text{HCF}(x - r, y - r, z - r)$
- ▶ Least number, remainder r each time: $\text{Required number} = \text{LCM}(x, y, z) + r$

09 Percentages

BASICS

- ▶ Definition: $x \% = x / 100$
- ▶ Fraction to percent: a/b as a percent $= (a/b) \times 100 \%$

INCREASE / DECREASE

- ▶ Increase %: $(\text{Increase} / \text{Original}) \times 100$
- ▶ Decrease %: $(\text{Decrease} / \text{Original}) \times 100$
- ▶ Value after $r\%$ increase: $\text{Original} \times (100 + r)/100$
- ▶ Value after $r\%$ decrease: $\text{Original} \times (100 - r)/100$

A VS B COMPARISONS

- ▶ A is $R\%$ more than B: B is less than A by $[R / (100 + R)] \times 100 \%$
- ▶ A is $R\%$ less than B: B is more than A by $[R / (100 - R)] \times 100 \%$

SUCCESSIVE % CHANGE

- ▶ Net effect of $x\%$ then $y\%$: $\text{Net change} = x + y + (xy/100) \%$ (use negative sign for a decrease)

10 Allegations and Mixtures

RULE OF ALLIGATION

- Cheaper : Dearer ratio: $(\text{Qty of cheaper}) / (\text{Qty of dearer}) = (\text{CP of dearer} - \text{Mean price}) / (\text{Mean price} - \text{CP of cheaper})$

MEAN PRICE

- Formula: $\text{Mean price} = (\text{Cost}_1 \times \text{Qty}_1 + \text{Cost}_2 \times \text{Qty}_2) / (\text{Qty}_1 + \text{Qty}_2)$

REPEATED REPLACEMENT

- After n dilutions with pure liquid: $\text{Remaining original quantity} = \text{Initial} \times [1 - (\text{replaced qty} / \text{total qty})]^n$

11 Probability

BASICS

- ▶ Probability of event E: $P(E) = (\text{favorable outcomes}) / (\text{total outcomes}), 0 \leq P(E) \leq 1$
- ▶ Complement: $P(\text{not } E) = 1 - P(E)$

COMBINED EVENTS

- ▶ Union (addition theorem): $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- ▶ Mutually exclusive events: $P(A \cup B) = P(A) + P(B)$
- ▶ Independent events: $P(A \cap B) = P(A) \times P(B)$

COUNTING TOOLS

- ▶ Permutations: ${}^n P_r = n! / (n - r)!$
- ▶ Combinations: ${}^n C_r = n! / [r!(n - r)!]$
- ▶ Symmetry: ${}^n C_r = {}^n C_{(n-r)}$

12 Ratios

BASICS

- ▶ Ratio: $a : b = a / b$ ($b \neq 0$)
- ▶ Scaling: $a : b$ is equivalent to $ka : kb$ for any $k \neq 0$

COMBINING RATIOS

- ▶ $a:b$ and $b:c \rightarrow a:b:c$: Scale both ratios so the common term (b) matches, then combine

COMPARING & MANIPULATING

- ▶ Comparing: $a:b > c:d \Leftrightarrow ad > bc$ (positive terms)
- ▶ Componendo-Dividendo: if $a/b = c/d$ then $(a+b)/(a-b) = (c+d)/(c-d)$

DIVIDING A QUANTITY

- ▶ Split N in ratio $a:b:c$: $\text{Parts} = N \times a/(a+b+c), N \times b/(a+b+c), N \times c/(a+b+c)$

13 Proportion

DIRECT & INVERSE PROPORTION

- ▶ Direct proportion: $x / y = k$ (constant) – both increase or decrease together
- ▶ Inverse proportion: $x \times y = k$ (constant) – one increases as the other decreases

PROPORTION RULE

- ▶ $a:b :: c:d$: $a \times d = b \times c$ (product of extremes = product of means)

MEAN & CONTINUED PROPORTION

- ▶ Mean proportional of a, b : $\sqrt{ab}$
- ▶ Continued proportion $a:b = b:c$: $b^2 = ac$

COMPOUND PROPORTION

- ▶ Chain rule: Multiply corresponding ratio terms together (used in time-work-wage chain problems)

14 Reasoning

NUMBER SERIES

- ▶ Pattern check order: Constant difference (AP) → constant ratio (GP) → difference-of-differences → alternating pattern → squares/cubes $\pm$ constant

CODING-DECODING

- ▶ Letter-to-number code: A=1, B=2, ... Z=26 (reverse code: Z=1 ... A=26)
- ▶ Shift cipher: Add or subtract a fixed offset to each letter's position in the alphabet

DIRECTION SENSE

- ▶ Net displacement (right-angle turns): distance = $\sqrt{x^2 + y^2}$, x = net East–West, y = net North–South shift

BLOOD RELATIONS & SYLLOGISMS

- ▶ Approach: No fixed formula – sketch a family tree / Venn diagram and eliminate options systematically

15 Work and Time

BASIC WORK RATE

- ▶ 1-day work: If A finishes work in n days, A's 1-day work = $1/n$
- ▶ Work done: $\text{Work} = \text{Rate} \times \text{Time}$

COMBINED WORK

- ▶ A and B together: $\text{Time together} = AB / (A + B)$ days (A, B = days each takes alone)
- ▶ Combined 1-day work: $1/A + 1/B$ (+ $1/C + \dots$)

EFFICIENCY

- ▶ A is k times as efficient as B: Ratio of time taken, $A : B = 1 : k$

PIPES & CISTERNS

- ▶ Sign convention: Inlet pipe = positive rate (fills) · Outlet pipe = negative rate (empties)
- ▶ Net rate (pipes together): Sum of individual signed rates
- ▶ Time to fill/empty: $1 / (\text{net rate})$

16 Geometry

TRIANGLES & ANGLES

- ▶ Angle sum of a triangle: 180°
- ▶ Exterior angle: = sum of the two non-adjacent interior angles
- ▶ Pythagoras theorem: $\text{hypotenuse}^2 = \text{base}^2 + \text{height}^2$ (right triangle)

CIRCLE THEOREMS

- ▶ Angle in a semicircle: 90°
- ▶ Angle at center vs circumference: Angle at center = $2 \times$ angle at circumference (same arc)
- ▶ Tangent property: Tangent is perpendicular to the radius at the point of contact

3D MENSURATION — VOLUME & SURFACE AREA

- ▶ Cube: $V = a^3$ · $TSA = 6a^2$ · Diagonal = $a\sqrt{3}$
- ▶ Cuboid: $V = lbh$ · $TSA = 2(lb + bh + hl)$
- ▶ Cylinder: $V = \pi r^2 h$ · $CSA = 2\pi rh$ · $TSA = 2\pi r(r + h)$
- ▶ Cone ($l = \sqrt{r^2 + h^2}$): $V = (1/3)\pi r^2 h$ · $CSA = \pi rl$ · $TSA = \pi r(r + l)$
- ▶ Sphere: $V = (4/3)\pi r^3$ · Surface area = $4\pi r^2$
- ▶ Hemisphere: $V = (2/3)\pi r^3$ · $CSA = 2\pi r^2$ · $TSA = 3\pi r^2$

17 Divisibility

DIVISIBILITY RULES

- ▶ By 2: Last digit is even (0, 2, 4, 6, 8)
- ▶ By 3: Sum of digits divisible by 3
- ▶ By 4: Last two digits divisible by 4
- ▶ By 5: Last digit is 0 or 5
- ▶ By 6: Divisible by both 2 and 3
- ▶ By 7: Double the last digit, subtract from the rest; repeat until the result's divisibility by 7 is clear
- ▶ By 8: Last three digits divisible by 8
- ▶ By 9: Sum of digits divisible by 9
- ▶ By 10: Last digit is 0
- ▶ By 11: (sum of digits at odd places – sum at even places) is 0 or divisible by 11
- ▶ By 12: Divisible by both 3 and 4
- ▶ By 13: Add four times the last digit to the rest; repeat until divisibility by 13 is clear

18 Profit and Loss

BASICS

- ▶ Profit: $\text{Profit} = \text{SP} - \text{CP}$ (when $\text{SP} > \text{CP}$)
- ▶ Loss: $\text{Loss} = \text{CP} - \text{SP}$ (when $\text{CP} > \text{SP}$)
- ▶ Profit / Loss %: $\text{Profit\%} = (\text{Profit}/\text{CP}) \times 100$ · $\text{Loss\%} = (\text{Loss}/\text{CP}) \times 100$

SP / CP FROM PERCENTAGES

- ▶ SP given profit%: $\text{SP} = \text{CP} \times (100 + \text{Profit\%}) / 100$
- ▶ SP given loss%: $\text{SP} = \text{CP} \times (100 - \text{Loss\%}) / 100$
- ▶ CP given profit%: $\text{CP} = \text{SP} \times 100 / (100 + \text{Profit\%})$
- ▶ CP given loss%: $\text{CP} = \text{SP} \times 100 / (100 - \text{Loss\%})$

MARKED PRICE & DISCOUNT

- ▶ Discount: $\text{Discount} = \text{MP} - \text{SP}$
- ▶ Discount %: $(\text{Discount} / \text{MP}) \times 100$
- ▶ SP from MP & discount%: $\text{SP} = \text{MP} \times (100 - \text{Discount\%}) / 100$

FALSE WEIGHT TRICK

- ▶ Profit % using a false weight: $[(\text{True weight} / \text{False weight}) - 1] \times 100$

19 Ages

SETTING UP THE EQUATION

- ▶ Variables: Let present age = $x \rightarrow n$ years ago = $x - n \rightarrow n$ years hence = $x + n$
- ▶ Method: Translate each clue (ratio, sum, difference) into a linear equation in x , then solve

RATIO SHORTCUT

- ▶ Ages in ratio $a:b$ now: After t years, ratio becomes $(ax + t) : (bx + t)$ – use the new condition to solve for x

20 Speed Distance and Time

BASICS

- ▶ Core relation: $\text{Speed} = \text{Distance} / \text{Time}$ · $\text{Distance} = \text{Speed} \times \text{Time}$ · $\text{Time} = \text{Distance} / \text{Speed}$
- ▶ Unit conversion: $1 \text{ km/hr} = 5/18 \text{ m/s}$ · $1 \text{ m/s} = 18/5 \text{ km/hr}$

AVERAGE SPEED

- ▶ Equal distances, speeds x, y : $2xy / (x + y)$
- ▶ Equal times, speeds x, y : $(x + y) / 2$

RELATIVE SPEED

- ▶ Same direction: $|s_1 - s_2|$
- ▶ Opposite directions: $s_1 + s_2$

TRAINS

- ▶ Crossing a pole / point: $\text{Time} = (\text{length of train}) / \text{speed}$
- ▶ Crossing a platform/bridge: $\text{Time} = (\text{length of train} + \text{length of platform}) / \text{speed}$
- ▶ Two trains crossing each other: $\text{Time} = (\text{sum of lengths}) / (\text{relative speed})$

BOATS & STREAMS

- ▶ Downstream / upstream speed: $\text{Downstream} = \text{boat} + \text{stream}$ · $\text{Upstream} = \text{boat} - \text{stream}$
- ▶ Boat / stream speed: $\text{Boat speed} = \frac{1}{2}(\text{down} + \text{up})$ · $\text{Stream speed} = \frac{1}{2}(\text{down} - \text{up})$

21 Pie Charts

READING A PIE CHART

- ▶ Central angle of a component: $(\text{Value of component} / \text{Total value}) \times 360^\circ$
- ▶ Percentage of a component: $(\text{Value of component} / \text{Total value}) \times 100$

REVERSE CALCULATIONS

- ▶ Value from angle: $(\text{Central angle} / 360^\circ) \times \text{Total value}$
- ▶ Value from percentage: $(\text{Percentage} / 100) \times \text{Total value}$

COMPARING COMPONENTS

- ▶ Ratio of two components: = ratio of their central angles = ratio of their percentages

22 Mean, Median, Mode, Variance and Standard Deviation

MEAN

- ▶ Ungrouped data: $\text{Mean} = \Sigma x / n$
- ▶ Grouped (frequency) data: $\text{Mean} = \Sigma(f \times x) / \Sigma f$

MEDIAN

- ▶ Ungrouped, n odd: $\text{Median} = \text{value of the } ((n+1)/2)\text{th term}$
- ▶ Ungrouped, n even: $\text{Median} = \text{average of the } (n/2)\text{th and } (n/2 + 1)\text{th terms}$
- ▶ Grouped data: $\text{Median} = L + [(n/2 - cf) / f] \times h$ (L = lower boundary of median class, cf = cumulative freq. before it, f = class frequency, h = class width)

MODE

- ▶ Ungrouped data: The most frequently occurring value
- ▶ Grouped data: $\text{Mode} = L + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$ (f_1 = modal class freq., f_0 = preceding, f_2 = succeeding)

SPREAD

- ▶ Range: $\text{Maximum} - \text{Minimum}$
- ▶ Variance: $\sigma^2 = \Sigma(x - \text{mean})^2 / n$
- ▶ Standard deviation: $\sigma = \sqrt{\text{Variance}}$

EMPIRICAL RELATION

- ▶ Mode-Median-Mean: $\text{Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean}$